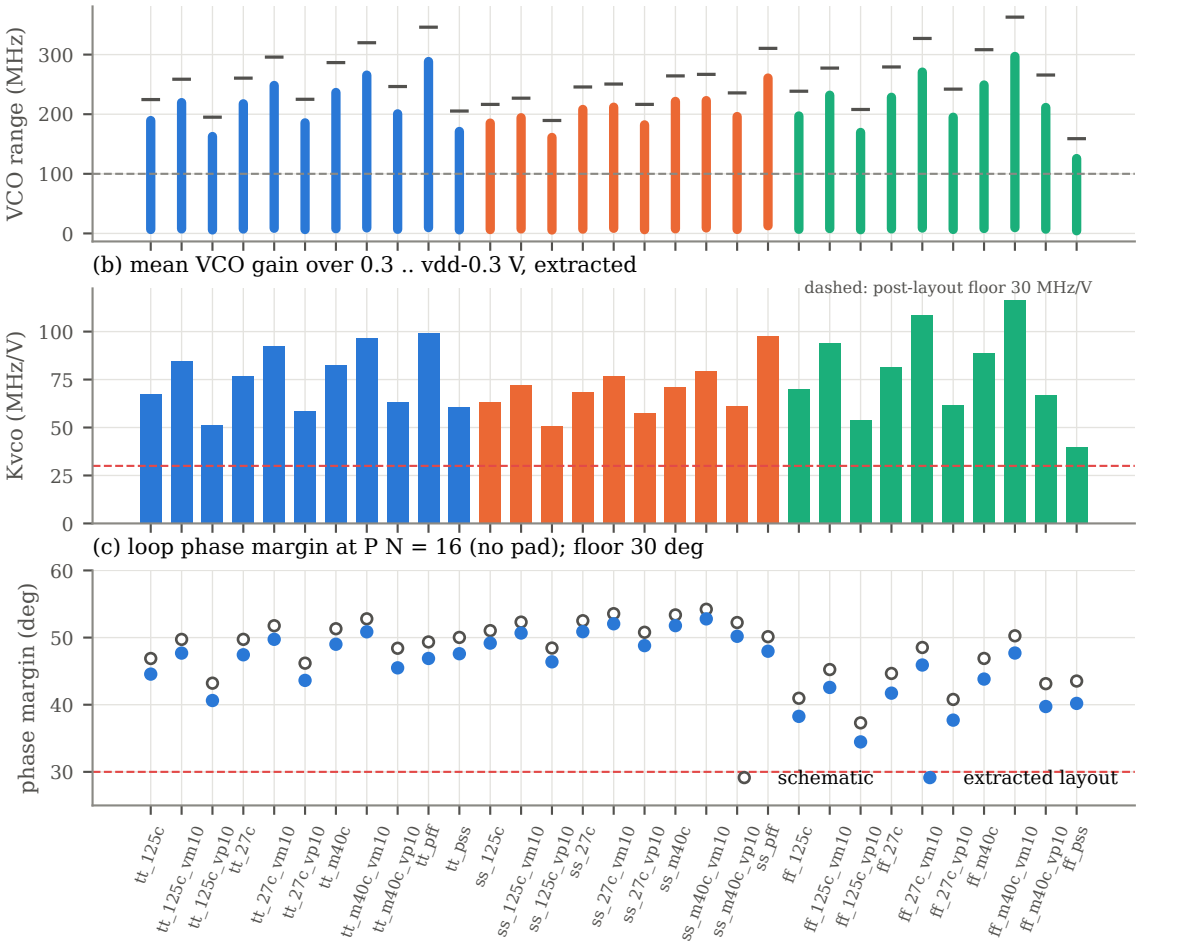
(a) post-layout VCO range, f(vctrl = 0.3 V) to f(vctrl = vdd); tick: pre-layout f(vdd); dashed: 100 MHz
VCO range (MHz)
0
100
200
300
(b) mean VCO gain over 0.3 .. vdd-0.3 V, extracted
dashed: post-layout floor 30 MHz/V
Kvco (MHz/V)
0
25
50
75
100
(c) loop phase margin at P N = 16 (no pad); floor 30 deg
phase margin (deg)
30
40
50
60
schematic
extracted layout
tt_125c
tt_125c_vm10
tt_125c_vp10
tt_27c
tt_27c_vm10
tt_27c_vp10
tt_m40c
tt_m40c_vm10
tt_m40c_vp10
tt_pff
tt_pss
ss_125c
ss_125c_vm10
ss_125c_vp10
ss_27c
ss_27c_vm10
ss_27c_vp10
ss_m40c
ss_m40c_vm10
ss_m40c_vp10
ss_pff
ff_125c
ff_125c_vm10
ff_125c_vp10
ff_27c
ff_27c_vm10
ff_27c_vp10
ff_m40c
ff_m40c_vm10
ff_m40c_vp10
ff_pss